

SKILL (Week 2)

Session -1

Task1: To Install Linux VM, Configure GCC, Setup Git Repository, Create Project Structure, Understand Shell Architecture, Build Initial Makefile.

```
varshini@DESKTOP-BLT5L3R:~$ tree
```

```
.
├── OSSP
│   ├── LICENSE
│   ├── Makefile
│   ├── README.md
│   ├── assets
│   │   └── architecture.png
│   ├── bin
│   │   └── my_shell
│   ├── docs
│   │   ├── design.md
│   │   └── report.pdf
│   ├── fork_exec
│   ├── fork_exec.c
│   ├── fork_exec.c.save
│   ├── include
│   │   ├── builtin.h
│   │   ├── executor.h
│   │   ├── parser.h
│   │   └── shell.h
│   ├── obj
│   ├── scripts
│   │   └── run.sh
│   ├── src
│   │   ├── builtin.c
│   │   ├── executor.c
│   │   ├── main.c
│   │   ├── parser.c
│   │   └── shell.c
│   └── tests
│       ├── test_executor.c
│       └── test_parser.c
├── fork_exec.c
├── hello
└── hello.c
```

Observation: I explored the Linux development environment and practiced organizing a shell project into appropriate directories and files. I also learned the role of GCC in compiling C programs and understood how Git helps in maintaining and tracking the project files. This activity helped me develop a clear understanding of how a software project is structured and managed systematically.

Task 2:

To Analyse process abstraction, execute fork(), understand exec() family, analyse parent-child relationships, inspect process tree, and practice system call tracing.

```
#include <stdio.h>
#include <stdlib.h>
#include <unistd.h>
#include <sys/wait.h>

int main()
{
    char command[100];

    printf("Enter a Linux command: ");
    scanf("%s", command);

    pid_t pid = fork();

    if(pid < 0)
    {
        printf("Fork failed!\n");
    }
    else if(pid == 0)
    {
        printf("Child Process\n");
        printf("Child PID: %d\n", getpid());

        execlp(command, command, NULL);

        printf("Execution failed\n");
    }
    else
    {
        printf("Parent Process\n");
        printf("Parent PID: %d\n", getpid());

        wait(NULL);

        printf("Child execution completed\n");
    }

    return 0;
}
```

```
varshini@DESKTOP-BLT5L3R:~$ cd ~/OSSP
varshini@DESKTOP-BLT5L3R:~/OSSP$ nano fork_exec.c
varshini@DESKTOP-BLT5L3R:~/OSSP$ gcc fork_exec.c -o fork_exec
varshini@DESKTOP-BLT5L3R:~/OSSP$ ./fork_exec
Enter a Linux command: ls
Parent Process
Parent PID: 942
Child Process
Child PID: 943
LICENSE  README.md  bin    fork_exec  fork_exec.c.save  obj    src
Makefile assets    docs  fork_exec.c  include          scripts tests
Child execution completed
varshini@DESKTOP-BLT5L3R:~/OSSP$ |
```